Introduction

A wind power company uses GIS to design turbine locations on a wind field. After creating a regular lattice, a tool is needed to adjust each turbine based on the local terrain. A graduate student studying environmental impacts of forest fires needs a matrix of triangular polygon features as a sampler. He needs a GIS tool to generate polygon features, which allows him to control the shape, size, spacing, alignment, and orientation of the features. A transportation researcher models work trips of individual workers who use public transport. However, the current GIS software does not support schedule-based network analysis. These are a few examples for which GIS programming is needed.

This book introduces ArcGIS Desktop add-in development using ArcObjects and Visual Basic .NET (VB.NET). An ArcGIS Desktop add-in is a software program that runs in an ArcGIS application such as ArcMap and ArcCatalog. An add-in can have a graphical user interface such as a button that performs tasks when clicked, or a tool that allows the user to interact with the map. An add-in may be a combo box that reacts to user selection, or a dockable window that contains multiple windows controls for complex tasks. An ArcGIS Desktop add-in project lets you create your own specialized extensions like ArcGIS built-in extensions as well as your own toolbars that hold add-in components and built-in commands.

ArcObjects is a set of platform-independent software components that make up ArcGIS applications. ArcObjects components are the building blocks of ArcGIS. Developers can program with these components to create custom GIS tools, functions, and even comprehensive GIS applications. ArcObjects components let developers access the native data objects and functions of ArcGIS. Programming with ArcObjects allows for defining of new data objects or features which support custom behaviors, therefore supports development of advanced applications that implement custom data models and algorithms.

Visual Basic .NET (VB.NET) is a high-level programming language implemented on the .NET Framework from Microsoft. Compared with other programming languages, the programming semantics and syntax of VB.NET are more resemble to natural human languages, making the language more readable and easier to learn. Moreover, the VB.NET Integrated Development Environment (IDE) provides rich and convenient tools for creating Graphical User Interface (GUI). The VB.NET IDE greatly enhances the efficiency of graphical user interface design. Furthermore, the VB.NET IDE provides powerful debugging tools that allow programs to be executed step-by-step so that programmers can examine variable values while a program runs. VB.NET debugging tools are not only effective in identifying and fixing errors, they are also extremely helpful for beginners to visualize program flows and understand programming logic. These features make VB.NET a good choice as the first programming language for beginners.

Readers of this book are expected to know GIS principles and be effective in using ArcGIS software. They are not required to have previous programming experience. The book uses six chapters to lay a foundation of VB.NET programming by introducing fundamental concepts and training skills necessary for ArcGIS Desktop add-in programming. These chapters cover Windows Graphical User Interface design, data types, variables, math operations, relational and logical operations, flow controls, arrays, object-oriented programming, as well as debugging skills, etc.

Readers who are comfortable with VB.NET programming may start with Chapter 7 which gets started with building ArcGIS Desktop add-in user interface elements. Chapter 8 introduces the ArcObjects library and explains how to interpret object model diagrams. Chapters 9 through 13 progressively present a series of ArcObjects programming techniques including loading data into ArcMap, accessing map layers, accessing features in feature classes, and manipulating spatial and attribute data. Chapter 14 deals with raster data and raster-based modeling and Chapter 15 introduces using geoprocessing tools and automating geoprocessing in add-in projects. As a capstone, Chapter 16 develops a Data Driven Pages add-in application which integrates all ArcObjects and add-in programming techniques and skills covered in this book.

To program with ArcObjects components, being able to interpret object model diagrams and to understand component interfaces and relationships between components are essential. While introducing a new component, tailed object model diagrams of related components are analyzed and discussed. Based on object model analysis, sample code snippets are provided to demonstrate capabilities and usage of the components.

Each chapter has a set of hands-on examples that let you develop workable projects or add-in tools. As you are instructed step by step to complete the examples, all code is provided to you. Key techniques are explained and discussed in the text and also explained as comments in the code. Please follow instructions carefully and make sure you understand what you are doing at each step. Many of the code blocks and procedures in the book are long, especially in later chapters. Be very careful in writing code when you follow the examples as a simple typo may take hours to debug.

The book uses the following icons to indicate different instructions.

 = “Do it”. This icon indicates tasks you must do to complete a sample project.

 = “Read it”. This icon is often used to indicate code demonstrating a technique. You do not need to enter the code in a program.

 = “For Your Information”. Additional information for you to know.

 = “Important”.

The CD-ROM accompanying the book contains data needed for the examples and exercises, sample programs presented in the book, code and image resources, as well as answers to exercise questions.

To successfully complete the examples and exercises in the book, you must have access to a Windows PC installed with ArcGIS Desktop 10.2, ArcObjects SDK 10.2, and Visual Studio 2010 (which must include Visual Basic). To learn VB.NET programming in Chapters 1 through 6, you may use Visual Basic 2010 Express that can be downloaded from Microsoft for free. Starting from Chapter 7, you must have ArcGIS Desktop and ArcObjects SDK on your computer.

To successfully complete the book, readers must be familiar with the Windows file system and be efficient in using Windows Explorer to manage directories (folders) and files. When working with the exercises in the book, you are suggested to

- Avoid using spaces in directory and file names. If you do need a space, use the underscore character (_) instead.
- Avoid using the following Microsoft reserved characters in directory and file names.
  < (less than) ? (question mark) “ (double quote mark)
  > (greater than) / (forward slash) : (colon)
  * (asterisk) \ (backward slash) | (vertical bar)
- Configure your Windows Explorer to display file name extensions. See how to change folder options at: http://windows.microsoft.com/en-us/windows7/change-folder-options.